Z-transform

Causality of Systems

Ex $h[n] = a^n u[n+1]$ $\forall n<0$

$h[n] \neq 0$ ≡ System is not causal.

Time domain inference :

Z-domain inference :?

$h[n] = a^n u[n]$

$h[n] = 0 \quad n<0$

Causal

$H(z) = \frac{1}{1-az^{-1}}; \quad |z|>|a|$

⇓ pole at $z=a$

① ROC is outside the outermost pole.

② $H(z)$ is rational in z

③ there is no pole at $z=\infty$

④ degree of $N(z) \leq$ degree of $D(z)$

Observations

1. Rational fn. in z
2. Two zeros at $z=0$
3. Two poles — $z=a$, $z=\infty$
4. ROC: $|z|>|a|$

∵ ∄ pole at $z=\infty$ ROC is not strictly outside the outermost pole

5. degree $N(z)$ > degree $D(z)$

$$h[n] = a^n u[n+1] = \frac{1}{a}\left\{a^{n+1} u[n+1]\right\}$$

$$x[n] \rightleftharpoons X(z)$$

$$x[n+1] \rightleftharpoons z\,X(z)$$

$$H(z) = \sum_{n=-1}^{\infty} a^n z^{-n}$$

$$= \sum_{n=0}^{\infty} a^n z^{-n} + a^{-1}z + a^{-1}z$$

$$= \frac{1}{1-az^{-1}} + a^{-1}z$$

$$= \frac{1 + a^{-1}z - 1}{1-az^{-1}} = \frac{a^{-1}z}{1-az^{-1}}$$

$$x[n] = a^n u[n] \rightleftharpoons \frac{1}{1-az^{-1}}$$

$$h[n] = \frac{1}{a}x[n+1] \rightleftharpoons \frac{1}{a}\cdot\frac{z}{1-az^{-1}}$$

$$= \frac{a^{-1}z}{1-az^{-1}}$$

$$H(z) = \frac{a^{-1}z^2}{z-a}$$

Ex

$$h[n] = \begin{cases} a^n & n \geq 0 \\ \frac{1}{(-n)!} & n < 0 \end{cases}$$

$$h[n] = h_1[n] + h_2[n]$$

where $h_1[n] = a^n u[n]$

& $h_2[n] = \begin{cases} 0 & n \geq 0 \\ \frac{1}{(-n)!} & n < 0 \end{cases}$

$H_1(z) = \frac{1}{1-az^{-1}}$

ROC: $|z| > |a|$

$$H_2(z) = \sum_{n=-1}^{-\infty} \frac{1}{(-n)!} z^{-n}$$

$$= \sum_{n=1}^{\infty} \frac{1}{n!} z^{n} + 1 - 1$$

$$H_2(z) = \sum_{n=0}^{\infty} \frac{z^n}{n!} - 1$$

$$= e^z - 1$$

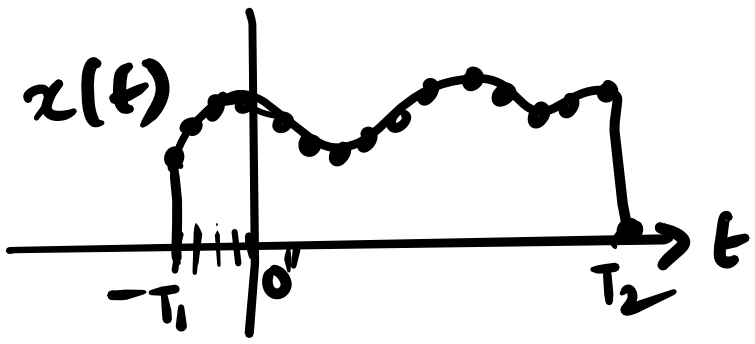
$$H(z) = \frac{1}{1-az^{-1}} + e^z - 1$$

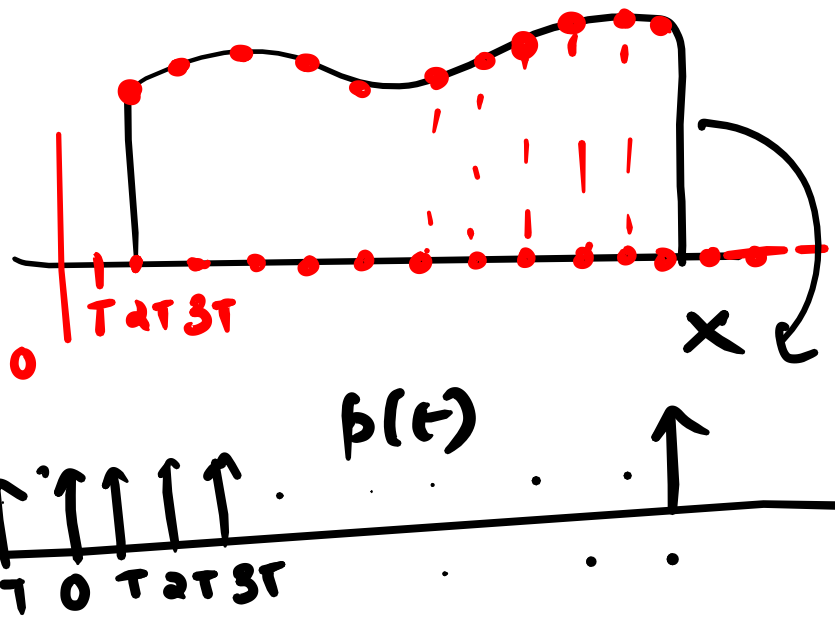
$\Downarrow$

$H(z)$ is not restricted in z

$\Rightarrow$ Not a causal system

ROC is $|z| > |a|$ but should exclude $z = 0$.



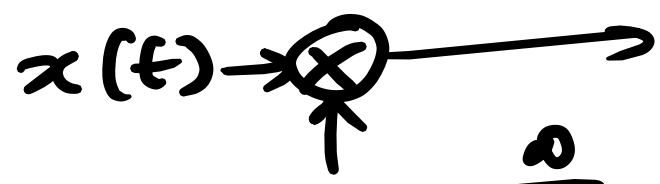


Samplig

$x(t)$ —Goal→ $x[n]$

$x[n] \approx x(nT)$

$T \equiv$ Samplig time

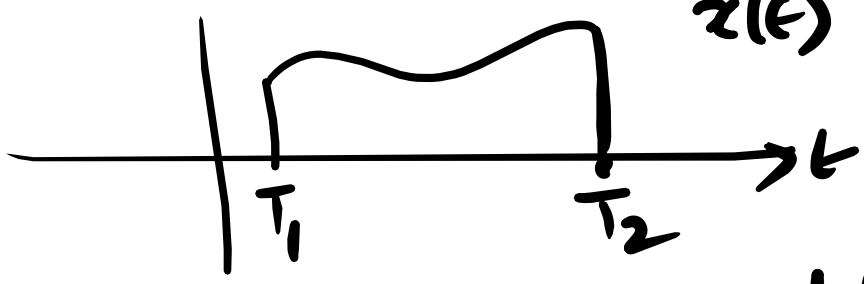


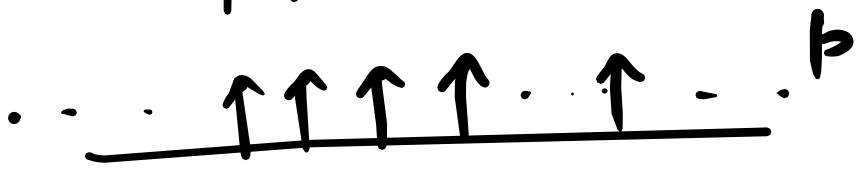
$$p(t) = \sum_{n=-\infty}^{\infty} \delta(t-nT)$$

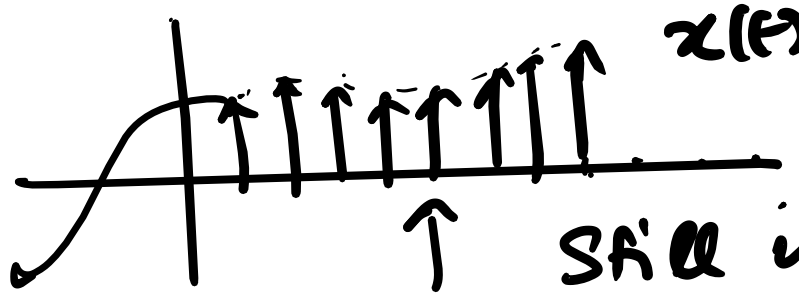
$$\hat{x}(t) = x(t)p(t)$$

$$= x(t)\sum_{n=-\infty}^{\infty}\delta(t-nT)$$

$$= \sum_{n=-\infty}^{\infty} x(nT)\delta(t-nT)$$



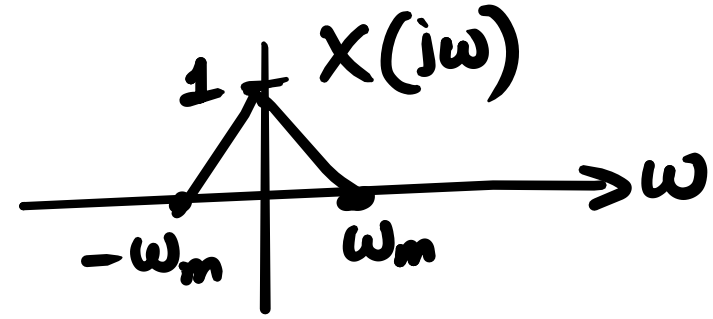




Still in continuous time domain —

$x(nT)\,\delta(t-nT)$

Consider $x(t)$ to be a band limited signal.



$X(j\omega) = 0 \;;\; |\omega| > \omega_m$

$$x(t)\,p(t) \rightleftharpoons \frac{1}{2\pi}\left[X(j\omega) * P(j\omega)\right]$$

$$p(t) = \sum_{n=-\infty}^{\infty} \delta(t-nT)$$

$$= \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

$$= \frac{1}{T}\sum_{k=-\infty}^{\infty} e^{jk\frac{2\pi}{T}t}$$

$$p(t) = \sum_{n=-\infty}^{\infty} \delta(t-nT) = \frac{1}{T}\sum_{k=-\infty}^{\infty} e^{jk\frac{2\pi}{T}t}$$

Let us assume

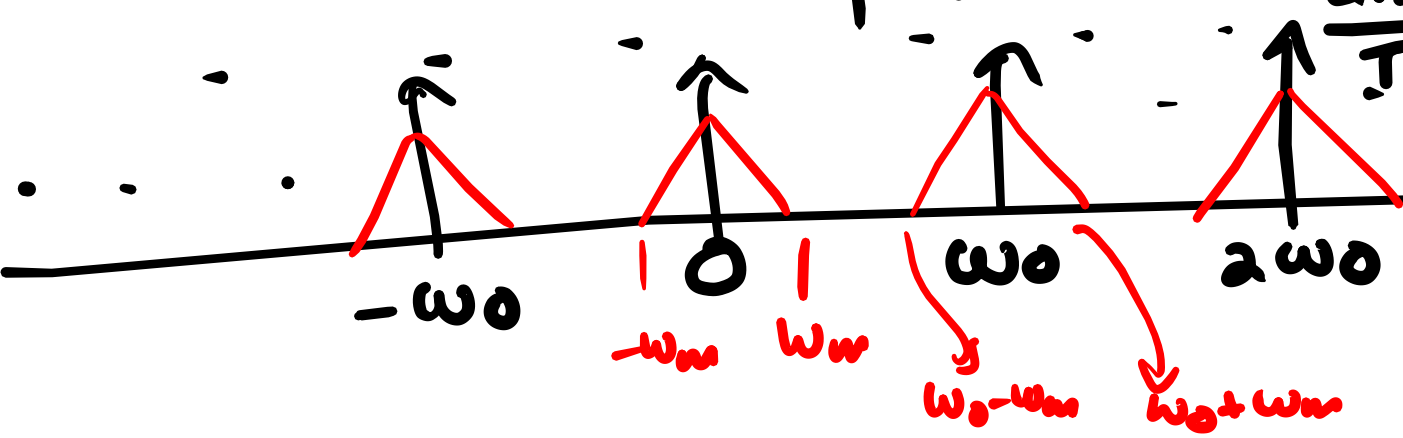
$\omega_m << \omega_0$

$$1 \xrightarrow{F.T} 2\pi\,\delta(\omega)$$

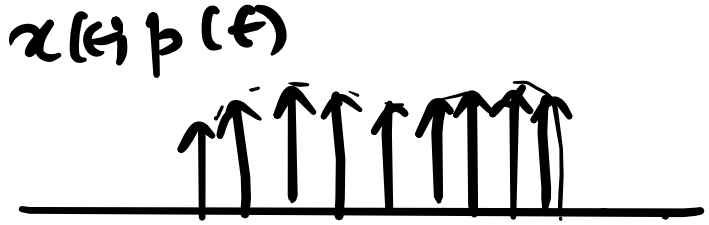
$$\sum_{k=-\infty}^{\infty} \frac{1}{T} e^{jk\omega_0 t} \rightleftharpoons \sum_{k=-\infty}^{\infty} \frac{2\pi}{T}\,\delta(\omega-k\omega_0)$$

$$\frac{1}{2\pi}\{X(j\omega) * P(j\omega)\} = ?$$

$$P(j\omega) = \frac{2\pi}{T}\sum_{k=-\infty}^{\infty} \delta(\omega-k\omega_0)$$



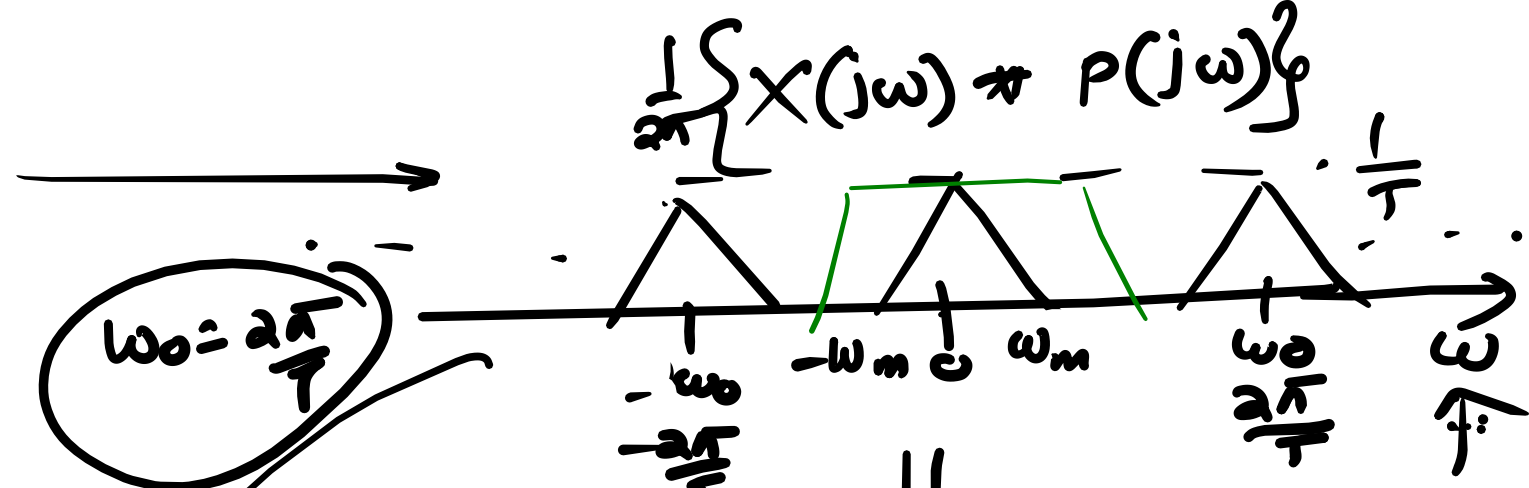
where
$\omega_0 - \omega_m = \omega_m$
or, $\omega_0 = 2\omega_m$
the two successive clips of spectrum touch each other



$$x(nT) \approx x[n]$$

Scale the frequency axis by T & correspondingly inversely scale in the time domain

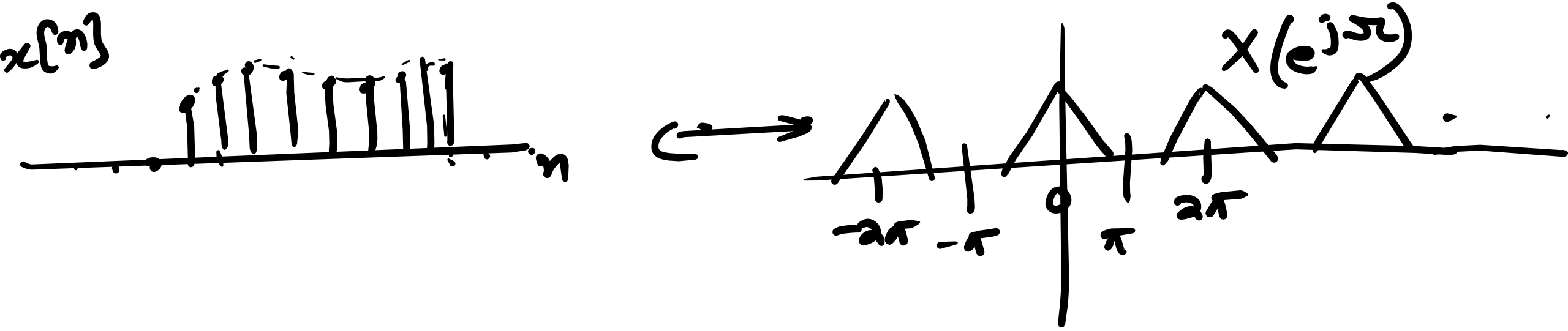
move to discrete time domain



In the continuous time domain
$\omega \equiv$ rad/s. (Units)

In the D.T. domain
$\Omega \equiv$ rad (Units)

$\Omega = \omega T \quad | \quad \omega = \Omega / T$


x[n]
n
X(e^{jΩ})
-2π
-π
0
π
2π